Haotian Liu

haotianliu.me@gmail.com | Homepage | GitHub | Google Scholar



EDUCATION

Xiamen University (XMU), Xiamen, China Sep. 2023 – Jun. 2027 (expected)

B.Eng. in Software Engineering | Overall Average: 82.52/100

Relevant Coursework:
Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); The Practice of Computer Programming (92).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents.
IJCNN 2026, **accepted**. Co-first author. arXiv:2604.08987

LFUSE: Learned Evidence Fusion for Noisy Spoken and Image-Grounded Product Search.
ICASSP 2027, under review. First author.

RESEARCH PROJECTS

CETI: Before the Agent Acts Mar. 2026 – Present

First Author | Joint research with Jialong Li | ZGC Lab

- Proposed CETI, a repository-grounded pre-coding review workflow that represents task constraints as editable **Records** linked to pinned source passages, allowing developers to revise an agent's task interpretation before code generation.
- Evaluated the workflow with 24 participants across 72 repository-task cases; the two-cycle Records condition showed the largest descriptive understanding gain (+17.2 percentage points adjusted contrast vs. FreeChat-2; HC3 95% CI [–6.0, 40.5]).
- Traced 102 confirmed request occurrences through 93 within-trajectory families, linking 48 behavioral families to patches and test evidence across 504 coding attempts on 10 tasks and 3 models.

Long-Horizon Memory Governance for LLM Agents (LHMG) Dec. 2025 – May 2026

First Author | Remote Research Collaboration | School of Data Science, Fudan University

- Developed a conflict-aware memory-governance framework for long-horizon LLM agents, combining revision graphs, No-Silent-Overwrite updates, risk-gated conditioning, and governed forgetting.
- Observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory in controlled diagnostics.

PilotBench: General Aviation Agent Benchmark Dec. 2025 – Mar. 2026

Core Contributor | Remote Research Collaboration | CESS, Beihang University

- Co-developed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases; evaluated 41 models with Pilot-Score, a composite of regression accuracy and instruction following and constraint compliance.
- Identified a precision–controllability dichotomy: traditional forecasters reached MAE 7.01, while LLMs achieved 86–89% instruction-following at 11–14 MAE; performance degraded in high-workload flight phases.

NeuroMark: Multimodal Reliability Evaluation for Neuropsychiatry Sep. 2025 – May 2026

Research Contributor | Remote Research Collaboration | School of Data Science, Fudan University

- Contributed to a multimodal neuropsychiatry benchmark that jointly evaluates text-based clinical reasoning and brain-imaging interpretation across 2,379 clinical questions and six imaging tasks.
- Evaluated 20 LLMs and 16 VLMs; helped develop SCR, EA, DDI, and JTI metrics to measure constraint compliance, effective accuracy, difficulty sensitivity, and cross-modal reliability.

INDUSTRY EXPERIENCE

LFUSE: Learned Evidence Fusion for Procurement Search Nov. 2025 – Apr. 2026

First Author | H World Group

- Developed LFUSE for ProcuraClaw, a learned evidence-fusion ranker that reranks BM25 and dense candidates using rank, character, pinyin, and query-agreement features, with noise-aware training for corrupted and ASR queries.
- Improved MRR@10 over tuned convex fusion by 7.8% on average across three human-judged Chinese Multi-CPR domains while reducing fusion-stage latency by 80–100× versus LLM-based adaptive fusion; transferred the model to a 1,854-SKU, 175-request hotel catalog.

RESEARCH AND TECHNICAL SKILLS

AI and Programming: LLM agents and memory, multimodal learning, AI for software engineering and HCI, RAG and information retrieval, benchmark and evaluation design; Python, PyTorch, Hugging Face, Spring Boot, SQL, Java/Java EE, C/C++, Unity.
Research Methods and Tools: Controlled experiments and human studies, ablation and error analysis, academic writing and presentations; Git, Linux, Docker, and LaTeX.
Languages and Tests: Chinese (native); English (TOEFL iBT 90); Japanese (reading knowledge).